Multilingual Voice AI for Pilgrim Assistance: STT Reliability, Bias by Invisibility, and Ethical Deployment in Sacred Environments

Ayesha Fareed
Cipheron Systems, Karachi, Pakistan

Abstract

The deployment of voice AI assistants in multilingual, safety-critical environments presents challenges that extend beyond prompt engineering. Building on prior work that proposed the SACRED Framework through two proof-of-concept voice assistants deployed at Masjid al-Haram (Fareed, 2026), this study addresses two previously identified gaps: multilingual speech recognition reliability and deployment oversight. Three commercial speech-to-text providers were evaluated within a production voice orchestration platform. Structured language routing achieved reliable transcription for English, Urdu, and Bahasa Indonesia, while Arabic remained unresolvable across four configurations and eight test calls. A structured field level evaluation across 41 calls in Urdu and Bahasa Indonesia found that identification numbers, the field most directly tied to a pilgrim's identity, were correctly relayed to staff in only 15 percent of Urdu calls and 52 percent of Bahasa Indonesia calls at final confirmation, with many of these errors traced specifically to the assistant's response generation rather than to speech recognition itself. A supplementary check using a second speaker produced results in a similar range, offering limited but direct evidence that this unreliability is not solely an artifact of one individual's accent or pronunciation. Cross-referencing language priorities against Hajj demographic data revealed that Bahasa Indonesia, representing the largest single national pilgrim quota of 221,000 pilgrims, was entirely omitted from the original design. This paper conceptualizes this as bias by invisibility, distinct from technical exclusion, and proposes empirical demographic verification as its remedy. Concrete ethical mechanisms, including a data retention policy, safety-critical escalation paths, and a spiritual distress acknowledgment mechanism, are proposed and tested within Shifa. A three-layer governance framework is introduced. Together these findings motivate four revisions to the SACRED Framework and the introduction of Oversight as a seventh principle, extending the framework to SACREDO.

Keywords

Multilingual Speech Recognition, Bias by Invisibility, AI Governance, Conversational AI, Cultural and Religious Sensitivity, Masjid al-Haram, Voice AI, SACREDO Framework, Human Oversight, Speech-to-Text Reliability

1. Introduction

Voice AI assistants are increasingly deployed in contexts where failure carries direct consequences for human safety. Among the most demanding of such contexts is Masjid

al-Haram in Makkah, Saudi Arabia, which receives millions of pilgrims annually from over 180 countries, representing extraordinary linguistic and cultural diversity within a safety-critical setting. A prior study applied prompt engineering to design two proof-of-concept conversational AI assistants for this environment (Fareed, 2026). The first, Noor, assists pilgrims who become lost or separated from their group by collecting name, identification, and location information through structured conversation and escalating it to relevant staff. The second, Shifa, performs medical emergency triage, gathering symptoms, pain level, and location data from distressed pilgrims for immediate dispatch to medical personnel. Both were built using persona design, priority ordering, constraint setting, and few-shot logic, deployed on the VAPI voice AI platform, and tested through simulated interactions. From that work, the SACRED Framework was proposed, a six-principle design guide covering Sensitivity, Accessibility, Clarity, Reliability, Escalation, and Debugging, intended as a reusable guide for AI practitioners working in culturally and religiously sensitive environments.

That study, however, operated in English only and identified two critical gaps requiring empirical follow-up: the absence of multilingual support, which limits accessibility for the majority of pilgrims whose primary languages include Arabic, Urdu, and Bahasa Indonesia; and the absence of a structured governance and oversight mechanism for deployment accountability. This paper addresses both gaps through systematic empirical investigation.

Three research questions guide this study. First, which commercial speech-to-text providers, as integrated within a production voice orchestration platform, can reliably support the primary pilgrim languages? Second, what ethical design mechanisms are required for safe and culturally appropriate deployment of AI triage assistants in sacred environments? Third, how should governance and oversight be structured for AI systems operating in high-stakes religious contexts?

The findings reveal systematic discrepancies between documented provider capabilities and real-world performance. Arabic failed across four independently tested configurations. A more significant finding emerged from demographic analysis: Bahasa Indonesia, spoken by pilgrims from Indonesia, which holds the single largest national Hajj quota globally, was never considered in the original design. This paper conceptualizes this as bias by invisibility, a distinct failure mode from technical exclusion, and proposes an empirical demographic verification step as its remedy. Alongside these findings, concrete ethical mechanisms are proposed and tested within Shifa, and a three-layer governance framework is introduced. Together these findings motivate four revisions to the SACRED Framework and the introduction of Oversight as a seventh principle, extending the framework to SACREDO.

This paper is organized as follows. Section 2 provides background on the prior work this study extends. Section 3 reviews relevant literature. Section 4 describes the methodology. Sections 5 and 6 present findings on STT reliability and bias by invisibility. Section 7 addresses ethical deployment mechanisms. Section 8 presents implications. Section 9 introduces the SACREDO Framework. Sections 10 and 11 address limitations and future work. Section 12 concludes.

2. Background

This paper extends a prior study that applied prompt engineering to design two conversational AI assistants for Masjid al-Haram (Fareed, 2026). That study introduced Noor, a lost pilgrim guidance assistant, and Shifa, a medical emergency triage assistant, both deployed on the VAPI voice AI platform using GPT-4.1 Mini as the language model and Deepgram for speech transcription. The assistants were built using persona design, priority ordering, constraint setting, and few-shot logic and were tested exclusively through simulated interactions within VAPI's voice simulation environment.

From that practice-based work, the SACRED Framework was proposed as a six-principle design guide: Sensitivity, embedding cultural and religious tone throughout; Accessibility, designing for diverse users and abilities; Clarity, structuring conversation to minimize cognitive load; Reliability, achieving consistent behavior through explicit priority ordering and override rules; Escalation, routing all emergencies to human staff without independent resolution; and Debugging, systematically isolating and resolving prompt-level conflicts. The study identified two critical limitations that directly motivate this paper: English-only language support, representing a significant accessibility gap for the majority of pilgrims, and the absence of any formal governance or oversight structure for deployment accountability.

3. Literature Review

The performance gap between high-resource and low-resource languages in automatic speech recognition is well documented and directly relevant to this study's empirical findings. Bian et al. (2025) demonstrate that this disparity persists even among widely spoken languages, finding that synthetic voice diversity techniques can partially mitigate but not eliminate the underlying training data imbalances that drive it. Chen et al. (2026) extend this through systematic benchmarking across multiple language families, concluding that low-resource languages remain critically under-evaluated for real-world deployment. This study empirically confirms both findings in a specific production context: Urdu required explicit provider configuration to achieve reliable transcription, and Arabic failed entirely across four tested configurations, consistent with the structural gaps both studies identify.

Beyond raw performance, ASR bias research examines systematic disparities affecting particular demographic groups. A systematic review by Nakatumba-Nabende et al. (2025) identifies imbalanced training data as the primary driver of these disparities and notes that privacy and reliability concerns are compounded when ASR is applied in sensitive domains such as medical contexts. This compound risk is directly relevant to Shifa's deployment environment, where transcription failures during medical triage are not merely inconvenient but potentially life-threatening.

Kuhlman et al. (2020) locate the bias problem upstream of technical implementation, arguing that a lack of diversity among AI practitioners shapes which populations are prioritized during system design, independent of any technical constraint. This observation grounds the distinction this paper proposes between bias by exclusion and bias by invisibility: the former originates in technical limitations, while the latter originates in the design-stage blind spots Kuhlman et al. identify. Ceccon et al. (2025) extend this framework toward regulatory compliance, proposing data bias profiles as an auditing tool, an approach whose underlying

logic this study applies operationally through the proposed demographic verification step.

The governance literature establishes that technical correctness alone is insufficient for safety-critical AI deployment. Alsayed (2026) proposes structured governance as a mechanism for managing deployment risk, a formalization of the oversight gap this study identifies through ad hoc transcript review. Legal scholarship has sharpened this further: Htin (2026) argues that distinct monitoring duties must be explicitly assigned across developers, deployers, and governing institutions. Healthcare-specific guidance reaches a parallel conclusion, with the Counsel Health Editorial Team (2026), summarizing CHAI's guidelines, holding that human involvement must be structured and meaningful in high-stakes care decisions. This study extends both arguments to a context neither addresses: AI deployment within a site of major religious significance, where accountability must incorporate religious and civil authority alongside technical expertise.

While ASR reliability, representation bias, and AI governance have each been independently studied, no prior work applies all three to conversational AI in a religious pilgrimage context, where language accessibility, cultural sensitivity, and safety-critical emergency response converge simultaneously. This paper addresses that gap using Noor and Shifa as a grounded case study.

4. Methodology

4.1 Research Approach

This study employs the same practice-based research methodology established in Fareed (2026): the design, testing, and iterative debugging of live voice assistants on the VAPI platform serve as the empirical foundation for the findings and framework revisions reported here. Rather than applying a predetermined theoretical model, the methodology evolved through hands-on configuration, live testing, and systematic fault investigation. Each phase directly informed subsequent design choices.

4.2 STT Provider Testing Protocol

Three commercial speech-to-text providers integrated within VAPI, namely Deepgram, Gladia, and Google, were systematically evaluated for their ability to support Arabic, Urdu, and Bahasa Indonesia. For each language and configuration, pre-scripted test phrases were used to isolate three behaviors: transcription accuracy, language identification, and correct downstream conversational logic. Table 1 reports the number of test calls per configuration.

Table 1: STT Provider Testing Protocol

Language	Provider	Test Calls	Configuration and Outcome
Arabic	**Deepgram Nova**	2	Auto-detect failed
Arabic	**Deepgram Whisper**	2	Language set accepted, output unintelligible
Arabic	**Gladia**	2	Auto-detect failed
Arabic	**Google/Gemini**	2	Explicit API key pipeline failure
Urdu	**Deepgram Nova-3**	20	Language set to ur, working
Bahasa Indonesia	**Deepgram Nova-3**	21	Language set to id, working

4.3 Ethical Mechanism Testing Protocol

To evaluate Shifa's ethical functions, scenario-based test scripts were created in advance. Each script isolated a single mechanism, such as critical-symptom escalation, confusion handling, or spiritual distress acknowledgment, ensuring each could be verified independently before being combined with others. Where a mechanism produced unexpected behavior, isolation testing was used to identify the specific instruction causing the failure.

4.4 Transparency and Failure Reporting

This study treats full reporting of failed configurations as a core methodological commitment. All four Arabic transcription configurations are documented, including their specific failure modes, rather than reporting only the final outcome. A system serving pilgrims in genuine emergencies must be held to a higher standard of honest capability assessment, and researchers documenting such systems bear a corresponding obligation of transparency. All testing was conducted within VAPI's voice simulation environment. Live field deployment with actual pilgrims remains outside the scope of this study.

5. STT Provider Comparison

To establish a reliable multilingual architecture for Noor and Shifa, three commercial speech-to-text providers integrated within VAPI were systematically evaluated. The following subsections document the configurations tested, observed outcomes, and failure modes identified for each provider.

5.1 Deepgram

The initial configuration utilized Deepgram's Flux General English model with the language parameter explicitly locked to English, rendering non-English speech as unintelligible phonetic approximations. Deepgram's multilingual auto-detect mode includes Urdu among its ten supported languages but excludes Arabic entirely. When auto-detection was bypassed in favor of explicitly setting a single supported language per assistant instance, the pipeline achieved stable and reliable transcription for English, Urdu, and Bahasa Indonesia. This configuration trades, sacrificing dynamic detection for demonstrated operational reliability, informed the structured language-routing architecture adopted throughout this study.

The absence of Arabic from Deepgram's supported language set represents a hard constraint rather than a configuration issue. No prompt-level adjustment or workaround was able to resolve this gap.

Deepgram's Whisper model was also evaluated as an alternative configuration specifically for Arabic transcription. Unlike Nova, which rejected the Arabic language parameter outright at the configuration stage, Whisper accepted the language setting but failed functionally in practice, returning Arabic speech as unintelligible phonetic approximations or failing to transcribe entirely. This distinction matters: Nova's failure is a hard provider-level exclusion, while Whisper's failure resembles the documentation-to-production drift pattern documented elsewhere in this study for Gladia and Google, a configuration that appears valid but does

not function as expected. Together, these two distinct failure modes confirm that the Arabic gap is not specific to one Deepgram model but reflects a broader absence of usable Arabic support within Deepgram's platform as accessed through VAPI.

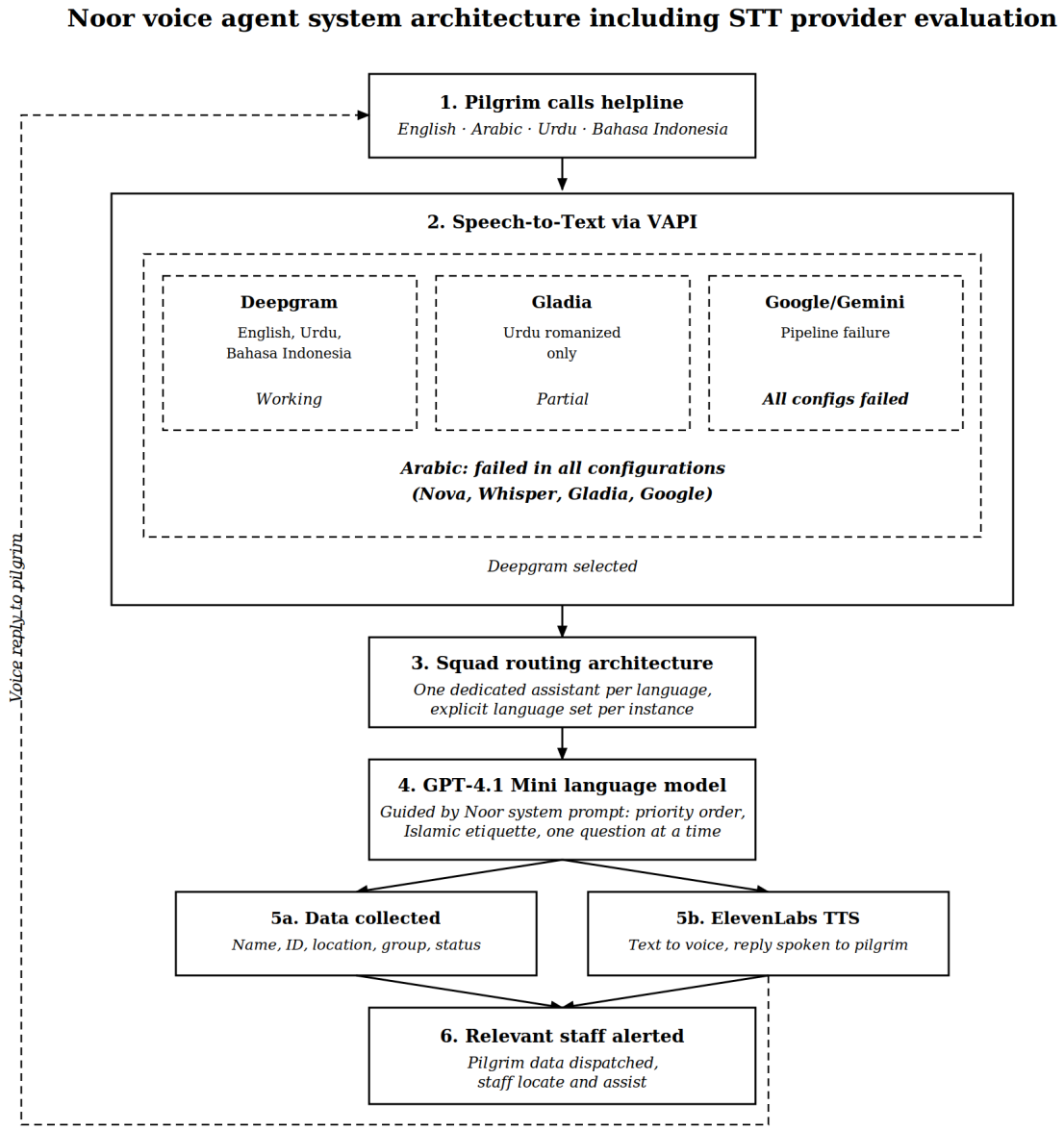


Figure 1: Noor voice agent system architecture including STT provider evaluation.

5.2 Gladia

Gladia was evaluated due to its advertised multilingual support and automatic language detection. Initial deployment of Gladia's latest model triggered a pipeline failure. Reverting to an older model restored stability but produced romanized Urdu script rather than native script, causing the downstream language model to misinterpret input as malformed English. A further issue emerged when a dashboard interaction silently overrode the automatic detection setting, requiring direct API-level intervention to correct. This silent override is documented as an instance of documentation-to-production drift, a failure mode where provider behavior diverges from documented behavior in ways not immediately visible to the developer.

5.3 Google

Investigation revealed that VAPI's Google provider routes audio through the Gemini multimodal model rather than Google's Cloud Speech-to-Text service, diverging from Google's own account setup wizard, which directs users toward Speech-to-Text credentials. Even after correctly provisioning a Gemini API key, test calls consistently resulted in pipeline failures. No audio was successfully transcribed across any tested configuration. This credential-routing mismatch represents a further instance of documentation-to-production drift, structurally related to but distinct from Gladia's silent override: in both cases, a developer following official documentation would not arrive at a working configuration.

5.4 Key Findings

Across all three providers, no configuration achieved reliable automatic language detection for the target multilingual profile. Table 2 summarizes the outcome for each configuration tested.

Table 2: STT Provider Comparison Summary

Provider	Languages Achieved	Primary Failure Mode
Deepgram explicit configuration	English, Urdu, Bahasa Indonesia	Arabic not supported in any configuration
Gladia	Urdu romanized only	Romanized script and silent settings override
Google/Gemini	None	Credential routing mismatch and pipeline failure

The findings reveal two distinct failure categories. The first is provider-level exclusion: Arabic is absent from Deepgram's supported language set entirely, and no configuration across Deepgram, Gladia, or Google successfully transcribed Arabic speech. The second is documentation-to-production drift: both Gladia and Google exhibited behavior that diverged from their documented capabilities in ways that would not be detectable without live testing in the specific integration environment.

Following the structured protocol described in Section 4.2, field-level accuracy was measured across all 20 Urdu calls and 21 Bahasa Indonesia calls. Table 2a reports the proportion of calls in which each field was correctly relayed in the assistant's final confirmation to staff, the value that would actually reach responding personnel in a real deployment.

Table 2a: Field Level Accuracy at Final Confirmation, by Language

Field	Urdu (n = 20)	Bahasa Indonesia (n = 21)
Pilgrim name	15/20 (75%)	12/21 (57%)
Identification number	3/20 (15%)	11/21 (52%)
Gate number	11/20 (55%)	19/21 (90%)

Three findings follow from this data. First, gate number showed the strongest reliability in both languages, though notably weaker for Urdu (55 percent) than for Bahasa Indonesia (90 percent). Second, the identification number was the least reliable field in both languages, correct in only 15 percent of Urdu calls and 52 percent of Bahasa Indonesia calls at final confirmation, despite the underlying digits frequently being transcribed correctly earlier in the same call. Third, this gap was investigated directly rather than left unexamined: two gate numbers, 65 and 92, were tested identically in both languages, allowing digit content to be held constant. For gate 65, Urdu accuracy was 40 percent against 60 percent for Bahasa Indonesia; for gate 92, Urdu accuracy was 40 percent against 100 percent for Bahasa Indonesia. Because the gap persists, and in the case of gate 92 widens, even when the exact digit sequence is identical across languages, the difficulty of the specific numbers selected can be ruled out as the explanation.

This data supports naming five distinct error types observed across testing. Digit omission describes a reference digit sequence losing one or more digits without replacement, the single most common identification number error observed. Digit substitution describes one digit replaced by a different digit. Digit duplication describes a partial or full repetition of the sequence, observed exclusively in Bahasa Indonesia testing. Confirmation stage corruption describes a value transcribed correctly earlier in a call being altered only when the assistant composes its final summary, indicating the error originates in generation rather than transcription. Field omission describes a required field, most seriously a pilgrim's name, being dropped entirely from the final confirmation with no value substituted.

5.5 Supplementary Robustness Check

A central limitation of the primary dataset, discussed further in Section 10, is that all 20 calls originated from a single speaker, making it impossible to distinguish provider-level reliability from speaker-specific reliability. To address this directly, a supplementary batch of 20 additional Urdu calls was conducted by a second, different speaker, using the same four phrase structures (name, identification number, gate number). This second speaker did not follow the original scripted names exactly, substituting alternative names across all four phrases (for example, "Nasreen" in place of "Ayesha Fareed"), so name level accuracy is not reported here, and because one call's confirmation was not fully captured, all percentages below exclude that single data point. This batch is reported as a separate, second speaker dataset rather than merged into the primary sample in Table 2a, preserving the integrity of the original controlled comparison while allowing a direct empirical check of whether the primary findings reflect a general system property or one speaker's particular accent and pronunciation.

Table 2b: Supplementary Robustness Check, Gate and Identification Number Accuracy

Metric	Original Dataset (n = 20)	Second Speaker (n = 20)
Gate number confirmed correct	11/20 (55%)	9/20 (45%)
Identification number confirmed correct	3/20 (15%)	4/20 (20%)

The second speaker's results were broadly consistent with the primary dataset, gate number confirmed correct in 45 percent of calls against 55 percent for the original speaker, and identification number confirmed correct in 20 percent of calls against 15 percent for the original speaker. This similarity is itself a meaningful finding: had the primary dataset's error rates been substantially specific to one speaker's accent or pronunciation, a second speaker would be expected to show a markedly different, likely improved, accuracy profile. Instead, both gate and identification number accuracy remained in a similar range, and both speakers showed the same underlying pattern of high spoken accuracy degrading at the confirmation stage. This provides direct, if limited, evidence that the unreliability documented in this study is not primarily an artifact of one individual's speech, though a sample of two speakers remains far short of what would be needed to generalize to the broader pilgrim population, as discussed in Section 10. Two further findings from this batch merit note. First, the digit substitution pattern converting gate number 65 to 56, originally documented in the primary dataset, recurred independently with this second speaker, corroborating that this is a specific, repeatable confusion rather than an artifact of one test session or one speaker. Second, this batch surfaced a failure mode not observed in the primary dataset: total collapse of a field into a non-numeric text fragment bearing no resemblance to the correct value, rather than a plausible but incorrect digit. This is reported as a preliminary observation warranting further investigation rather than a confirmed additional error category, given it was observed with only this one additional speaker.

The practical implication for deployment is significant. Any multilingual voice AI system built on this platform must currently be architected around explicit, per-language assistant instances rather than dynamic detection. This is a stronger constraint than existing ASR literature suggests, since that literature evaluates models in isolation rather than as integrated within commercial voice orchestration platforms. A developer relying solely on provider documentation would arrive at incorrect conclusions about both coverage and reliability.

6. Bias by Invisibility

Prior to this study, the prioritization of English, Urdu, and Arabic within Noor's architecture relied on informal assumptions of pilgrim demographics rather than empirical data. Cross-referencing these assumptions against 2025 Hajj quota data revealed a significant misalignment: Indonesia received the single largest national pilgrim quota, 221,000 pilgrims, exceeding any Arabic-speaking nation's allocation (Pakistan Point, 2025). This diversity is corroborated by Saudi Arabia's own Hajj Statistics Publication, which recorded pilgrims from

approximately 171 different nationalities in 2025 (GASTAT, 2025). Despite this, Bahasa Indonesia had been entirely omitted from initial design considerations and the Section 5 provider investigation.

This study distinguishes this finding from the Arabic technical gap, categorizing it under a separate taxonomy of algorithmic bias. Bias by exclusion describes a failure mode where a language is actively targeted but fails due to identifiable technical constraints: Arabic was attempted across four configurations and failed each time. Bias by invisibility describes a language omitted from the design phase entirely, not due to technical limitations, but because it fell outside the design team's frame of reference, despite representing the largest single affected population. The distinction matters because these two failure modes require different remedies. Bias by exclusion can in principle be resolved through better engineering. Bias by invisibility cannot be engineered away: it will recur in every future iteration unless the design pipeline explicitly mandates an empirical demographic verification step before language priorities are set.

Following this identification, Bahasa Indonesia was added to the test suite and confirmed as working via live testing using Deepgram Nova-3 with the language parameter explicitly set to Indonesian. Testing also surfaced a prompt conflict in which a hardcoded English string in the greeting overrode the system's general language instruction, a bug structurally identical to an issue documented in Fareed (2026). The recurrence of this bug class across two independent implementation efforts suggests it represents a systematic category of prompt engineering vulnerability rather than an isolated incident.

The humanitarian implication is direct: a pilgrim whose primary language is Bahasa Indonesia, calling Noor in an emergency, would have received no intelligible assistance from the original system. This is not a degraded experience but a complete failure of access to emergency aid for the single largest national pilgrim contingent at the world's most densely populated religious site.

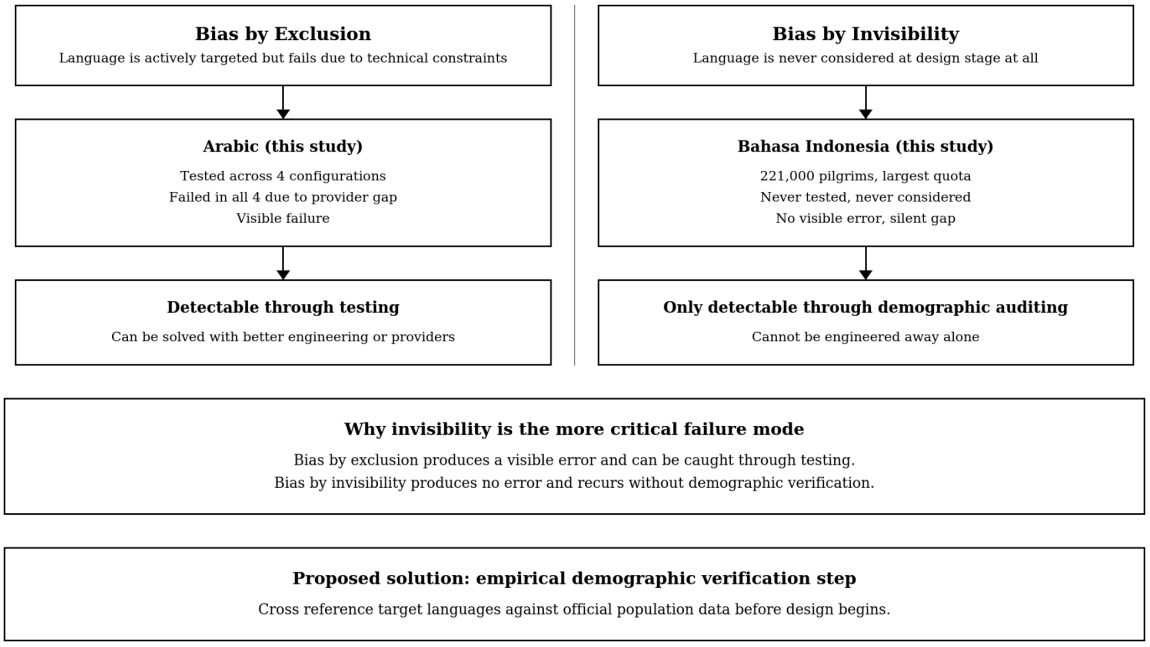


Figure 2: Bias by exclusion versus bias by invisibility in multilingual AI design

7. Ethical Deployment

In addition to multilingual accessibility, this study identifies four ethical gaps in the original design: data privacy and retention, safety-critical failure handling, cultural and spiritual sensitivity, and deployment oversight. Each is examined below, with concrete design mechanisms developed, implemented, and tested within Shifa's live configuration.

7.1 Privacy and Data Retention

Prior work noted that Noor and Shifa retain no data beyond an active call, which avoids persistent storage risk but creates a hazard: a dropped call mid-emergency loses all partially collected medical information. This study resolves that tension by implementing a data retention policy for Shifa anchored in the principles of purpose limitation and least-privilege access. Location data is purged immediately upon the arrival of responding personnel. Granular data, including symptoms, pain metrics, pilgrim names, and identification credentials, is retained long-term exclusively where a verified medical event occurred and received an active response. All data is systematically deleted in the event of a false alarm. Access is restricted solely to authorized medical personnel. This framework should be evaluated against Saudi Arabia's Personal Data Protection Law, which governs health information as a protected category, before any live deployment.

7.2 Safety-Critical Escalation

Two limitations were identified in Shifa's original design: the absence of a self-monitoring mechanism to detect comprehension failures and an invariant question sequence that failed to adapt to symptom severity despite the explicit definition of an immediate/Critical urgency tier. Two mechanisms were designed and tested in response. A confusion escalation rule halts the standard sequence if a pilgrim's responses are unclear for two consecutive turns, triggering a staff alert rather than looping indefinitely. This worked correctly on the first test. A critical symptom-fast-path bypasses the standard sequence upon initial detection of high-severity symptoms, requesting location immediately to enable physical dispatch and collecting name only afterward if the pilgrim remains responsive. Testing this fast path surfaced a distinct bug: the assistant's end-call function was invoked before the required closing phrase was spoken. This ordering conflict between text generation and function invocation was resolved only once the instruction explicitly sequenced the two action types as separate turns.

7.3 Spiritual Distress Acknowledgment

Shifa's original architecture modeled pilgrim distress along a single dimension restricted to physical pain, lacking the capacity to recognize spiritual distress such as an inability to perform ritual prayers. Two design approaches were evaluated and rejected: a comprehensive emotional support mode, which threatened to dilute Shifa's primary clinical purpose, and complete omission, which would violate the Sensitivity principle of the SACRED Framework. The resulting design introduced a constrained single-sentence acknowledgment: Shifa briefly recognizes spiritual distress before immediately returning to clinical triage, with subsequent non-medical inquiries strictly prohibited. Testing with

concurrent physical and spiritual disclosures confirmed that this mechanism delivers localized acknowledgment without conversational diversion.

7.4 Governance and Oversight

The vulnerabilities documented in this study were identified exclusively through ad hoc researcher intervention and manual transcript analysis. No structured protocol existed to detect operational failures during unmonitored deployment. To address this, this study proposes a tripartite oversight framework comprising three distinct layers: technical oversight, evaluating systemic alignment with intended functional specifications; content and domain oversight, ensuring clinical and theological appropriateness through expertise distinct from software engineering; and jurisdictional oversight, establishing formal accountability to governing religious and civil authorities. At present, only the technical layer operates, and purely on an informal basis. The pragmatically achievable interim measure proposed here is a transition from ad hoc discovery to a formalized lightweight protocol for periodic transcript reviews.

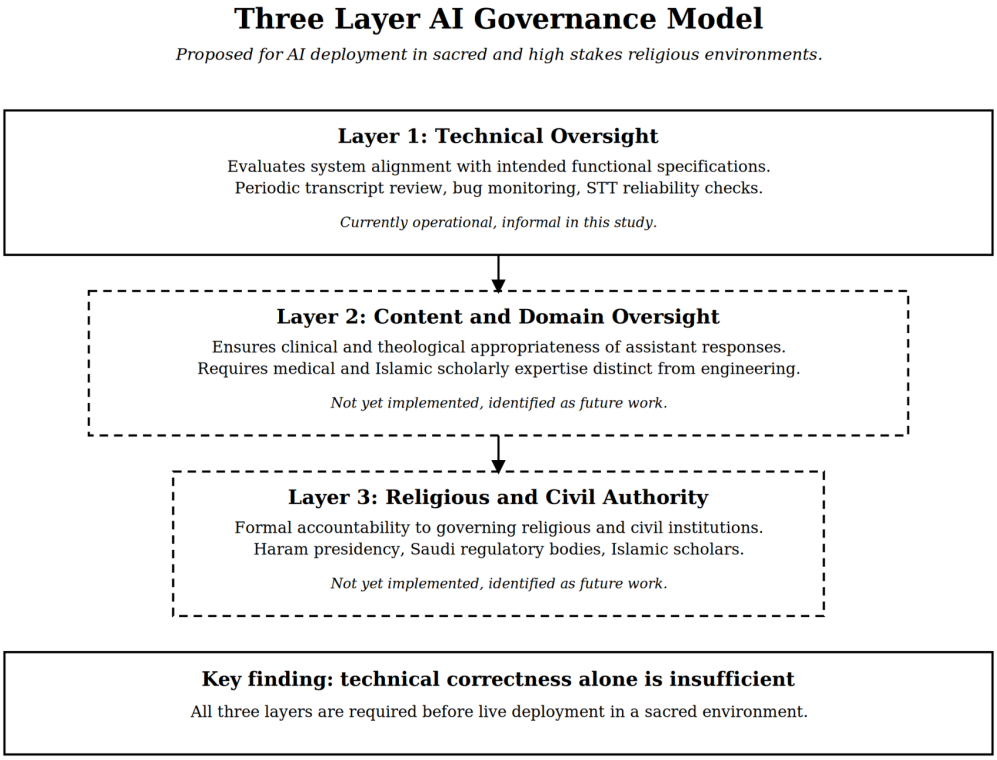


Figure 3: Proposed three layer governance model, the SACREDO Oversight principle.

8. Implications

8.1 Current Implications

The findings of this study carry immediate practical consequences for anyone designing or evaluating multilingual voice AI systems. The failure of automatic language detection across all three tested providers demonstrates that language support claims made by commercial STT providers cannot be taken at face value when those providers are accessed through a voice orchestration layer. Developers must empirically verify each language provider

combination in their specific integration environment before making accessibility claims about their system.

The concept of bias by invisibility has implications beyond this study's deployment context. Any AI system serving a demographically diverse population risks excluding its largest user groups not through technical failure but through design-stage omission. The proposed remedy, mandatory empirical demographic verification before language priorities are set, is transferable to any domain where a system's intended users span multiple linguistic communities.

The field level accuracy data in Section 5.4 carries a specific implication that is easy to overlook: many of the identification number errors documented in this study occurred not because the pilgrim's speech was mistranscribed, but because the language model corrupted an already correctly transcribed value while composing its closing summary to staff. This distinction matters for where developers should focus their testing effort. A system's speech-to-text accuracy and its ability to accurately relay already-correct information in a generated response are two separate failure points, and this study's data suggests the second is at least as significant a source of error as the first. Developers building similar systems should test both stages independently rather than assuming that accurate transcription guarantees an accurate final output.

The governance gap documented here is also not unique to sacred environments. The finding that all bugs in this study were caught through ad hoc researcher intervention rather than structured oversight applies to the majority of production AI systems deployed by small teams or individual practitioners, where formal oversight structures are rarely in place.

8.2 Future Implications

The SACREDO framework, if validated through independent application, could serve as a practical design standard for AI systems in any culturally sensitive, high-stakes environment beyond Islamic pilgrimage sites. The bias by invisibility concept, if adopted as a formal evaluation criterion, could inform how regulators and auditors assess AI accessibility claims under frameworks such as the EU AI Act.

The Arabic transcription failure documented across four configurations has implications for the STT industry. The absence of Arabic support in production-accessible configurations reflects a commercial prioritization that disproportionately affects one of the world's most widely spoken language communities. Systematic documentation and publication of findings of this kind may contribute to greater accountability in provider language support decisions.

The second speaker check reported in Section 5.5 points toward a further implication worth stating explicitly. Because a second speaker's results fell within a similar range to the primary speaker's, the reliability problems documented in this study are more plausibly a structural property of the underlying provider and model pipeline than something an individual pilgrim could avoid by speaking more clearly or slowly. If this pattern holds across a larger and more demographically varied speaker sample, as proposed in Section 11, it would strengthen the case that fixes belong at the system level, in provider selection, model configuration, and confirmation-stage validation, rather than being framed as a user-side problem.

The three-layer governance model proposed here offers a template for how AI accountability might be structured in contexts where technical, domain, and civil authority oversight are all legitimately required. As AI deployment in religious, medical, and other sensitive environments continues to expand, frameworks of this kind will become increasingly necessary.

9. SACREDO Framework

The empirical and ethical findings detailed in Sections 5 through 7 expose gaps that require theoretical evolution of the SACRED Framework proposed in Fareed (2026). This section proposes SACREDO as a provisional extension, revising four existing principles and introducing a seventh. Table 3 summarizes the full framework.

Table 3: SACREDO Framework Principles and Revisions

Principle	Original (SACRED)	Revision (SACREDO)
Sensitivity	Cultural and religious tone embedded in design	Extended to dynamic, real time acknowledgment of context-specific distress
Accessibility	Design for diverse users	Requires empirical demographic auditing; distinguishes bias by exclusion from bias by invisibility
Clarity	One question at a time; no jargon	Unchanged
Reliability	Consistent, predictable behavior	Reformulated to treat internal fault detection as an intrinsic property of reliability
Escalation	Route to human staff	Extended to adaptive, priority based routing informed by symptom severity
Debugging	Systematic prompt fault resolution	Unchanged
Oversight (new)	Not present in SACRED	Multi tiered accountability structure required for high stakes autonomous deployments

The Sensitivity principle is extended from static cultural tone to dynamic, runtime acknowledgment of context-specific distress, motivated by the spiritual distress acknowledgment mechanism developed for Shifa in Section 7.3. The Accessibility principle is revised to require empirical demographic auditing rather than assumed coverage, directly motivated by the bias by invisibility finding in Section 6. The Clarity and Debugging principles remain unchanged. The Reliability principle is reformulated to treat internal fault detection as an intrinsic reliability property rather than an external add-on, motivated by the confusion escalation mechanism documented in Section 7.2. The Escalation principle is extended to adaptive, priority-indexed routing based on symptom severity, motivated by the critical symptom fast path developed for Shifa in Section 7.2.

The seventh principle, Oversight, is introduced as a direct response to the governance gap documented in Section 7.4. It establishes that a multi-tiered accountability architecture is required for high-stakes autonomous deployments in sacred environments and that its absence is itself a documented failure mode rather than a minor gap.

SACREDO is posited here as a nascent framework open to further scholarly validation. Its seven principles are derived directly from empirical observation rather than proposed abstractly, consistent with the practice-based methodology established in Fareed (2026).

10. Limitations

This study inherits the foundational constraints of prior work: all evaluation was conducted within VAPI's voice simulation environment rather than through live field deployment with real pilgrims, and the research was executed by a single investigator without external validation of design decisions or testing outcomes.

Several constraints are specific to this study. Sample sizes per language configuration were as follows: Arabic was tested eight times across four provider configurations, two tests per configuration. Urdu was tested twenty times across four different test phrases using Deepgram Nova-3. Bahasa Indonesia was tested twenty times using Deepgram Nova-3, with one additional confirmation call conducted to verify a specific prompt conflict identified during testing, bringing the total to twenty-one calls. While these sample sizes are sufficient for diagnostic fault detection and pattern identification, they remain insufficient for statistically generalizable reliability claims. A single, targeted exception is the supplementary second speaker check reported in Section 5.5, conducted for Urdu only since a second Bahasa Indonesia speaking tester was not available during this study: a second speaker's results fell within a similar range to the primary Urdu speaker's, providing limited but direct evidence that the documented unreliability is not solely an artifact of one individual's accent or pronunciation. This remains a sample of two speakers, however, and does not substitute for the systematic, multi-speaker evaluation identified as necessary future work below.

A structured field-level evaluation was conducted for Urdu and Bahasa Indonesia, reported in Table 2a, though this remains a binary correct or incorrect judgment per field rather than a comprehensive word-level error count. The standard measure for ASR accuracy is Word Error Rate, defined as:

$$WER = (S + D + I) / N$$

where S is the number of substituted words, D is the number of deleted words, I is the number of inserted words, and N is the total number of words in the reference transcript. While a comprehensive WER computation across every field and call was not conducted, digit level WER was calculated directly for three verified reference and hypothesis pairs where an identification number was corrupted specifically during the assistant's final confirmation: two independent instances of the reference sequence 6-7-8-9-0 rendered as 6-7-8-1-0 (a single digit substitution, edit distance 1, WER 20 percent), and one instance of the reference sequence 9-1-3-5-7 rendered as 1-9-3-5-7 (a two digit transposition, treated under standard Levenshtein distance as two substitutions, edit distance 2, WER 40 percent). These figures corroborate the direction and approximate magnitude of the field level accuracy percentages reported in Table 2a, though they represent a small, targeted subsample rather than a comprehensive computation, and should not be averaged or compared numerically with the Table 2a percentages as though measuring the same thing. Future work should calculate WER across all tested configurations to enable quantitative comparison against published ASR benchmarks.

Arabic transcription failures remain unresolved despite four configurations and eight test calls. The conclusion that Arabic is fundamentally unsupported is based on empirical elimination rather than direct access to provider documentation or engineering support. The

proposed governance framework describes an idealized structure of which only one informal tier currently exists. SACREDO therefore remains unvalidated beyond Noor and Shifa, with the Oversight principle awaiting empirical implementation.

11. Future Work

Several directions follow directly from this study's limitations. Arabic transcription requires resolution, either through alternative providers, a custom pipeline independent of native platform integrations, or direct engagement with Deepgram regarding Arabic support roadmaps. This is the most urgent technical gap, since Arabic remains the official language of Saudi Arabia and a primary language for a significant proportion of Haram pilgrims.

Quantitative evaluation should be expanded. Larger test samples and formal WER calculations would move findings from diagnostic to statistically reliable, enabling meaningful comparison against published ASR benchmarks and strengthening the evidentiary basis for the provider conclusions reported here. Multi-speaker testing should also be expanded. The second-speaker check reported here was limited to Urdu; Bahasa Indonesia's reliability findings rest entirely on a single speaker, since no second Bahasa Indonesia-speaking tester was available during this study. Recruiting a demographically varied sample of speakers across all three languages, particularly Bahasa Indonesia, would confirm whether the reliability patterns identified in this study reflect a general system property rather than characteristics specific to individual speakers.

SACREDO should be applied to an independent deployment by a different practitioner to test its generalizability beyond Noor and Shifa. Until that occurs, it remains a single-source framework derived from one researcher's experience. The governance framework should also progress from proposal to practice: formalizing periodic transcript review as a routine procedure and establishing pathways toward content, domain, and religious authority oversight are achievable near-term steps that do not require institutional infrastructure to begin.

Finally, live field evaluation with actual pilgrims, conducted under appropriate institutional and ethical review, remains the primary empirical test of all findings reported here. Simulation-based testing can identify prompt-level conflicts but cannot replicate the unpredictability of real human conversation under genuine distress. Real-world deployment data would either validate or substantially revise the conclusions of both this study and its predecessor.

12. Conclusion

This study extends the SACRED Framework through empirical investigation of two gaps identified in prior work: multilingual speech recognition reliability and deployment oversight. The findings demonstrate that commercial STT providers, as currently integrated within production voice orchestration platforms, cannot be assumed to match their documented capabilities. Arabic failed across four configurations and eight test calls, and documentation-to-production drift was identified as a recurring failure class affecting both Gladia and Google. A structured field-level evaluation across 41 calls in Urdu and Bahasa Indonesia found that identification numbers, the field most directly tied to a pilgrim's identity, were correctly relayed to staff in only 15 percent of Urdu calls and 52 percent of Bahasa

Indonesia calls at final confirmation, with many of these errors traced specifically to the assistant's response generation rather than to speech recognition itself. A supplementary check using a second speaker produced results in a similar range, offering limited but direct evidence that this unreliability is not solely an artifact of one individual's accent or pronunciation. These are not theoretical concerns but documented, reproducible failures with direct consequences for any practitioner building multilingual voice AI on this platform.

The study also introduces bias by invisibility, evidenced by the complete omission of Bahasa Indonesia from the original design despite Indonesia holding the single largest national Hajj quota globally. This concept generalizes beyond this deployment: any AI system serving a linguistically diverse population is vulnerable to the same failure unless demographic verification is made an explicit design step.

Concrete ethical mechanisms, including a data retention policy, safety-critical escalation paths, a spiritual distress acknowledgment mechanism, and a three-layer governance framework, are proposed and empirically tested. Together these findings motivate four revisions to existing SACRED principles and the introduction of Oversight as a seventh principle, extending the framework to SACREDO.

Throughout, this study treats transparent reporting of failures as a core contribution. Systems designed for vulnerable populations in safety-critical environments require honest capability assessment. The limitations identified here are not weaknesses to be minimized but an honest accounting of what this study established and what remains to be investigated.

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